Discover: Enzyme Action: Preparing the pH Investigation

Today I can… □ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. □ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme’s work. □ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Lesson strip

Today I can

Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity.

Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme’s work.

Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

In your book

In your book: Glue in today's strip; date it. Answer the four clues.

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Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Supported

1. Think of a eukaryotic animal cell. Name the part of that cell which holds the chromosomes.

2. Link to today: Think about diffusion. Which way do the particles go? Do they go from a higher concentration to a lower one, or the reverse?

3. Think about osmosis. There is a partially permeable membrane. A substance goes through it. What is the name of that substance?

4. A transport process uses energy to move particles. They go against a concentration gradient. What is the name of that process?

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Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Exit Ticket · Supported

Evidence

Plan: find out how pH affects amylase as it acts on starch. Iodine turns blue-black when it detects starch. It stays orange-brown when it does not detect starch. Today we plan the test for later. We have not yet collected any results from a reaction.

1. Why can amylase act on starch and then act again? Use the model or complete: 'Starch fits the ___. The enzyme is not ___.'

Word bank: active site · used up · product. Choose the words that explain binding and reuse.

2. For the pH check, name what we change, what we time and one thing we keep the same.

Use the headings Change / Time until… / Keep the same. Use the detector key to finish the timing rule.

Under the heading Exit ticket: answer the short questions in your book.

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Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Independent

1. Think of a eukaryotic animal cell. Name the part of that cell which holds the chromosomes.

2. Link to today: Think about diffusion. Which way do the particles go? Do they go from a higher concentration to a lower one, or the reverse?

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Discover: Enzyme Action: Preparing the pH Investigation

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Exit Ticket · Independent

Evidence

Plan: find out how pH affects amylase as it acts on starch. Iodine turns blue-black when it detects starch. It stays orange-brown when it does not detect starch. Today we plan the test for later. We have not yet collected any results from a reaction.

1. Explain why amylase acts on starch and can be used again.

Link the fit at the active site to specificity. Show how the shapes fit each other. Then say how the enzyme and substrate differ.

2. Name the IV, define the timed DV with units, and name one control with a reason.

The DV needs an endpoint we can see. Explain how a change in another rate factor would affect a comparison of pH.

Under the heading Exit ticket: answer the short questions in your book.

Discover: Enzyme Action: Preparing the pH Investigation

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Arrival · Aspire

1. Think of a eukaryotic animal cell. Name the part of that cell which holds the chromosomes.

2. Link to today: Think about diffusion. Which way do the particles go? Do they go from a higher concentration to a lower one, or the reverse?

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Exit Ticket · Aspire

Evidence

Plan: find out how pH affects amylase as it acts on starch. Iodine turns blue-black when it detects starch. It stays orange-brown when it does not detect starch. Today we plan the test for later. We have not yet collected any results from a reaction.

1. Explain enzyme specificity. Explain how the enzyme can act again. Predict how an unsuitable pH could change the way this works.

Link the active-site shape to binding and to products leaving. Do not assume that every drop in activity proves permanent denaturation.

2. State the IV. State the endpoint-time DV and its units. Give one control to make the pH comparison fair. Justify your choice.

Use a stopping rule we can see. Show how your control rules out a different cause.

Under the heading Exit ticket: answer the short questions in your book.

Discover: Enzyme Action: Preparing the pH Investigation

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Starter and method strip

Starter

Which iodine observation shows starch is still present: blue-black or orange-brown?

Method strip

F · Set up the results headings. Do this before you start.

B · Make equal mixtures. Keep the temperature the same. Use the approved setup.

D · Mix enzyme with starch. Start timing as directed.

A · Take a sample at each fixed time gap. Put it in its own iodine drop.

C · Find the first sample that leaves iodine orange-brown. Record it.

E · Repeat at each pH. Keep the same conditions.

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

You Do · Independent

Write a testable prediction for how pH could affect amylase acting on starch. Explain why the enzyme can act again. [2 marks]

Name the IV and define the timed DV with units. Give three controls and justify one. [6 marks]

Order the six method cards and explain why iodine is used to test separate sampled drops. [2 marks]

Distinguish a no-active-amylase control sample from a control variable. Predict the control result. [3 marks]

State suitable table headings for pH, trial, endpoint time / s, observation and relative rate 1/time / s^{-1}. State what to record if there is no endpoint by 300 s. [3 marks]

I · S · C

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… □ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. □ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. □ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

You Do · Supported

1. Use the model to finish: "Amylase is the ___. Starch is the ___. The enzyme is not ___."

Word bank: enzyme · substrate · used up. Trace the purple shape before and after.

2. Match pH / time to endpoint / temperature to CHANGE / MEASURE / KEEP THE SAME. Give one reason for keeping temperature the same.

Independent = change. Dependent = measure. Temperature can also change reaction speed.

3. Order method cards A–F on the reference page. Point to the actions done before mixing.

Find BEFORE (F/B), DURING (D/A), then ENDPOINT and REPEATS (C/E).

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

You Do · Supported · continued

4. Point to the endpoint sample. Predict the iodine result in a separate sample with starch but no active amylase.

Blue-black = starch detected; orange-brown remains = not detected. The control still contains starch.

5. Use these results headings: pH | trial | endpoint time / s | observation. If no endpoint is seen by 300 s, choose "0 s" or "no endpoint by 300 s".

The agreed stopping time is a limit on the observation. It is not a measured reaction time.

I · S · C

Discover: Enzyme Action: Preparing the pH Investigation

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You Do · Choice

Choose the catalyst: amylase / starch / iodine. Predict: pH 4 will reach the endpoint sooner / later than pH 7. Give one active-site reason. Complete: The enzyme can act again because it is not ___. Record C beside your book response.

Match pH / endpoint time / temperature to change / measure / keep the same. Give one reason. Order the six supplied method cards. Choose the iodine check: separate sampled drops / the reaction mixture. Choose no active amylase as a comparison group / changed pH; predict blue-black / orange-brown.

Choose the change heading: pH / temperature; repeat heading: trial / colour; observation heading: colour / pH. Choose time unit: s / s^{-1}; rate unit: s^{-1} / s. Choose the truthful entry at 300 s: zero / no endpoint by 300 s. Give one reason.

After choosing, write one short scientific sentence, label or reason in your book.

I · S · C

Discover: Enzyme Action: Preparing the pH Investigation

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You Do · Aspire

Predict a pH–activity pattern that we can test. Explain specificity. Explain how the enzyme acts again. Do not claim that slower activity proves permanent denaturation. [4 marks]

Link how it works to a prediction we can measure. A shape model shows what might happen. It does not prove the optimum pH.

State the IV. Define a DV that someone else could measure in the same way. Give four controls. Justify two of them. Say what other cause each rules out. [8 marks]

Operational means someone else can use the same stopping rule. Temperature and concentration can each affect activity, even if pH stays the same.

Order method cards A–F. Explain why iodine goes in separate check drops. Explain why we must start in the same way. [3 marks]

Start timing with mixing in the same way each time. Do not let the detector become a second condition that changes.

Discover: Enzyme Action: Preparing the pH Investigation

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You Do · Aspire · continued

Show how a no-active-amylase control differs from a control variable. If its result is orange-brown when you did not expect that, what would you check? [3 marks]

Think back from the starch you expect to be there. First ask if an unexpected control makes the test valid. Do this before using it to explain how the enzyme works.

Use these example samples: blue-black at 90 s and orange-brown at 120 s. Samples were 30 s apart. State the time to record. State the time window when starch crossed the detection threshold. Give one way to make timing more precise. A further run has no endpoint by 300 s. Choose a truthful table entry. [4 marks]

The first sample with no starch detected gives the latest time bound. Choose these headings: pH / trial / time or no endpoint by… / observation. Explain what your improvement deals with.

I · S · C

Discover: Enzyme Action: Preparing the pH Investigation

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You Do · shared evidence

Evidence and reference diagrams

Enzyme: a biological catalyst. It speeds a reaction up. It is not used up. Substrate: the substance it acts on. pH: how acid or alkaline a substance is. Amylase acts on starch. It makes smaller sugars.

Iodine key: blue-black = starch detected. Orange-brown remains = starch not detected. check separate sampled drops. Endpoint = first sampled drop leaving iodine orange-brown; record its time / s.

Card | Action

F | Prepare the results headings before starting.

B | Use the setup your centre has approved. Prepare the mixtures so they match. Match their temperature too.

D | Mix enzyme and starch; start timing consistently as directed.

A | Take samples at the fixed interval. Put each sample in its own iodine drop.

C | Record the first sample where iodine remains orange-brown.

E | Repeat at each pH using matched conditions.

For a valid comparison, change pH. Keep temperature the same. Match enzyme and starch concentrations and volumes. Match the total volume, the start and how you take samples. A separate control with no active amylase should still test positive for starch. That control is not the same as a factor kept constant.

Discover: Enzyme Action: Preparing the pH Investigation

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You Do · shared evidence · continued

Evidence and reference diagrams · continued

This task is planning only. Use the centre-approved practical method later. Table headings: pH | trial | endpoint time / s | observation. To add a relative-rate column, 1/time / s^{-1}, you need an endpoint you have seen. The starting amounts must match.

This is an example: blue-black at 90 s; orange-brown at 120 s. Samples were 30 s apart. Record 120 s. The starch crossed the detection threshold after 90 s and by 120 s. A further run has no endpoint by 300 s. We cannot state its rate as a number.

Discover: Enzyme Action: Preparing the pH Investigation

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You Do · shared evidence · continued

Reference

Enzyme: a biological catalyst. It speeds a reaction up. It is not used up. Substrate: the substance it acts on. pH: how acid or alkaline a substance is. Amylase acts on starch. It makes smaller sugars.

starch substrate

amylase enzyme

smaller sugar products

active site

Discover: Enzyme Action: Preparing the pH Investigation

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Knowledge organiser

Learning goal

I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

A complementary active site

Specificity means an enzyme acts on particular substrates. Amylase breaks starch down into smaller sugars. It speeds this change up. The products leave. The enzyme is not used up. It can act again. This shape model is simple. It does not show real molecules as rigid shapes.

Connect pH to a measured endpoint

pH tells us how acid or alkaline a substance is. An unsuitable pH can change the active site. This can slow activity. Denaturation disrupts the enzyme's three-dimensional shape and how it works. Slow activity alone does not prove permanent denaturation.

Plan a valid comparison

Change pH. Time the first sampled drop that leaves iodine orange-brown. This means starch is not detected. Blue-black means starch is detected. Match temperature, concentrations, volumes and sampling. Check separate drops. Keep iodine out of the main mixture.

Discover: Enzyme Action: Preparing the pH Investigation

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Knowledge organiser · continued

Read the evidence honestly

A comparison with no active amylase should still test positive for starch. If no endpoint is seen by 300 s, write "no endpoint by 300 s". Do not make up a zero time or zero rate. The best pH is still to be found.

Diagram

Purple shows amylase. Yellow shows starch and smaller sugar products. The shape model shows two moments.

starch substrate

amylase enzyme

smaller sugar products

active site

Discover: Enzyme Action: Preparing the pH Investigation

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Word page

Enzyme and substrate

enzyme — A biological catalyst. It speeds up a reaction. It is not used up.

substrate — The substance that an enzyme acts on.

active site — The part of an enzyme where the substrate binds. The shapes fit each other.

specificity — An enzyme acts on particular substrates. They fit its active site.

denaturation — A change in enzyme shape. This stops the active site binding as it should.

Measure and compare

pH — A measure of how acid or alkaline a substance is.

independent variable — The factor we choose to change.

dependent variable — The outcome we measure.

control variable — A factor we keep the same. This makes the comparison fair.

endpoint — The agreed point we can see and record. We record the time when it is reached.

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

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Word page · continued

Diagram

Purple shows amylase. Yellow shows starch and smaller sugar products. The shape model shows two moments.

Specificity means an enzyme acts on particular substrates. Amylase breaks starch down into smaller sugars. It speeds this change up. The products leave. The enzyme is not used up. It can act again. This shape model is simple. It does not show real molecules as rigid shapes.

starch substrate

amylase enzyme

smaller sugar products

active site

Discover: Enzyme Action: Preparing the pH Investigation

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We Do · response cards

Response cards

A B C

Guided example

In your book: Complete the shared task; record one decision and its reason.

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… □ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. □ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. □ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

We Do · cards and recording

Use the cards

In your book: Complete the shared task; record one decision and its reason.

Choose a card, then a destination. You can change your mind.

Card 1

Smaller sugar products

Card 2

Active site

Card 3

Amylase enzyme

Card 4

Starch substrate

Discover: Enzyme Action: Preparing the pH Investigation

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Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

We Do · cards and recording · continued

Destinations

1 · Large purple shape

2 · Whole yellow piece

3 · Notch in purple shape

4 · Separated yellow pieces

In your book: Complete the shared task; record one decision and its reason.

Guided example

In your book: Record one evidence claim and explain its limit.

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

We Do · cards and recording

Use the cards

In your book: Complete the shared task; record one decision and its reason.

Choose a card, then a destination. You can change your mind.

Card 1

A · Take a sample at each fixed time gap. Put it in its own iodine drop.

Card 2

B · Make equal mixtures. Keep the temperature the same. Use the approved setup.

Card 3

C · Find the first sample that leaves iodine orange-brown. Record it.

Card 4

D · Mix enzyme with starch. Start timing as directed.

Card 5

E · Repeat at each pH. Keep the same conditions.

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme’s work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

We Do · cards and recording · continued

Card 6

F · Set up the results headings. Do this before you start.

Guided example

In your book: Record one evidence claim and explain its limit.

Discover: Enzyme Action: Preparing the pH Investigation

Today I can… □ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. □ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. □ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

I Do · readiness bridge · Supported

1. Catalyst and substrate

An enzyme is a biological catalyst: it speeds a reaction without being used up. Choose: speeds up / always stops a reaction.

The fact is in the first sentence.

2. Substrate or comparison

Amylase breaks down starch. Which is its substrate here: starch, protein or lipid?

Substrate means the substance the enzyme acts on.

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I Do · readiness bridge · Supported · continued

3. Controls or repeats

To check the effect of pH, should we change pH only or change pH and temperature together?

Keep other rate factors the same.

4. Changed factor and endpoint

We change pH in a test. We measure the time it takes. Name the independent variable. Is it pH or time?

Independent means the factor we choose to change in the test.

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I Do · readiness bridge · Independent

1. Catalyst and substrate

Starting fact: an enzyme is a biological catalyst. A catalyst speeds a reaction without being used up. Describe its job.

Use both parts of the supplied definition.

2. Substrate or comparison

Starting fact: amylase breaks starch into smaller sugars. Name the substrate and the enzyme.

The substrate is what is acted on.

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I Do · readiness bridge · Independent · continued

3. Controls or repeats

Why keep temperature the same when testing the effect of pH on a reaction?

Temperature can also affect reaction speed.

4. Changed factor and endpoint

We change pH in a test. We time how long it takes to reach the reaction endpoint. Name the independent variable. Name the dependent variable.

Which factor do we change? Which outcome do we measure?

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I Do · readiness bridge · Aspire

1. Catalyst and substrate

Starting fact: catalysts speed reactions without being used up. How is the role of amylase different from the role of starch?

Amylase is the enzyme; starch is its substrate.

2. Substrate or comparison

Two trials use different pH values. They also use different amylase concentrations. Why is that a problem?

Can you be sure that pH alone caused the effect?

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I Do · readiness bridge · Aspire · continued

3. Controls or repeats

Why repeat a reaction-time measurement at the same pH?

Think about variation and a doubtful result.

4. Changed factor and endpoint

We are going to compare times in a test. First, we must all agree what counts as the reaction endpoint. Why must we agree this before we start?

Think about the rule we use to stop timing. That rule is part of how we measure.